Selector-Threshold Matching, Kadanoff-Baym Evolution, and Wilson-Line Completion

Angus Muffatti

Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

40. First mixed selector-threshold topology

Using only the displayed interactions

$$Q^\dagger Q R^2, \qquad R H^\dagger H, \qquad H\bar{q}D, \qquad a\bar{D}D,$$

the first connected mixed graph has

$$I = 8, \qquad V = 6, \qquad L = I - V + 1 = 3.$$

It is one-particle irreducible but two-particle reducible. In the self-consistent 2PI description, it is generated by coupled lower-loop kernels rather than one primitive 2PI skeleton. The transient operator has

$$\Delta V^{(3)}_{Qa} = \epsilon C_3 \,\mathrm{Re}[e^{ia/f_a} \mathcal{Q}_1] + O(\epsilon^2).$$

A first NDA estimate made it negligible relative to the intended thermal potential. The subsequent exact factorised matching confirmed that conclusion.

41. Two-time surrogate and what it did not prove

The project evolved complete unequal-time propagators for an exact quadratic non-Markovian surrogate with 20 coordinates, 10 momentum modes, 1,801 time steps, and 301 stored two-time slices. It preserved the symplectic structure to 7.11×10^{-15}, reproduced $T_5/T_0 = 1/4$, and suppressed unselected leakage to approximately 1.14×10^{-7}.

This was a genuine two-time consistency test. It was not a nonlinear, non-Abelian $3 + 1$ dimensional plasma simulation. The distinction remains explicit [@Tranberg2024; @Lindner2007; @Bhattacharya2022].

42. Failure of the direct bosonic inflaton portal

The original trilinear

$$\mathcal{L}_{\phi R} = -\frac{g_{\phi R}}{2}\phi R^2$$

requires a coupling large enough that post-inflationary oscillations make selected and nominally heavy reheaton modes tachyonic. The illustrative resonance parameter was $q_{\rm end} \sim 6 \times 10^3$. Late-time kinematic closure therefore does not prevent early tachyonic or resonant production. This portal fails generically [@Dufaux2006].

43. Selector-gated fermionic cascade

Replace it with a complete fermionic parent orbit N_k:

$$\mathcal{L}_N = -y_\phi \phi \sum_k \bar{N}_k N_k - \sum_k m_{N_k}(Q) \bar{N}_k N_k - y_R \sum_k R_k \bar{N}_k \nu_k + \text{h.c.}$$

On the one-hot branch, N_0 is accessible while the other copies are extremely heavy. The hidden Landau-Zener exponent in the benchmark is about 460.4, giving a linear production suppression near 10^{-200}. Selected fermions can be nonperturbatively produced, but Pauli blocking caps their occupancy rather than allowing unbounded bosonic amplification [@GreeneKofman1998].

The repair passes at the resolved linear and cascade levels. Full backreaction, rescattering, and gauge-plasma production remain part of the nonlinear programme.

44. Deconstructed Wilson-line completion

A candidate quality completion uses an 18-link deconstructed gauge theory. The collective Wilson line is

$$\mathcal{W} = \prod_{j=0}^{17} \frac{\Sigma_j}{f/\sqrt{2}} = e^{ia/f_a}.$$

A cyclic transformation shifts replica labels and rotates $\mathcal{W}$. A messenger chain generates

$$\mathcal{L}_{\Psi,\text{eff}} = -\sum_k \left[M - \kappa(\omega^k \mathcal{W} + \omega^{-k} \mathcal{W}^\dagger)\right] \bar{\Psi}_k \Psi_k.$$

The first local winding operator requires all 18 links, so low-dimension local operators cannot directly depend on the complete phase. This is the type of nonlocal/collective protection needed to turn the infrared Z_6 model into a credible gauge-quality completion [@Hor2026].

Perturbative discrete anomaly sums pass for the displayed vectorlike orbits, but the full global/cobordism anomaly problem is open [@Byakti2017].